U.S. DEPARTMENT OF ENERGY'S
CYBERFORCE® PROGRAM

CyberForce® 101

CyberForce Competition

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CyberForce Competition® Overview

Purpose

The CyberForce Competition® is an in-person competition that provides collegiate students with a cyber-physical hands-on, realistic energy-based scenario to secure and defend that incorporates life-like constraints and tasks, actual end users, and continual emerging threats. Students are challenged by the opportunity to apply classroom knowledge in a real-life practical environment.

The goal is to build the pipeline of qualified cybersecurity candidates for crucial positions such as those in our energy sector, national laboratories, and industry. The CyberForce Competition provides an inclusive space for all participants, including collegiate students, mentors, industry, and academia.

Benefits

The competition provides an inclusive space for all participants, including collegiate students, mentors, industry, and academia. As an in-person event, there is the ability to network, enhance collegiate programs, and engage in mentorship from industry and national laboratory staff. Through targeted training on cybersecurity skill sets and introductions to examples of emerging trends and technologies, the CyberForce Competition provides a space to grown and learn.

Students are challenged by the opportunity to apply classroom knowledge in a real-life practical environment. This is simulated through red-team intrusions, real-time defense measures, communication with senior leadership, and more. Participating in the CyberForce Competition emphasizes knowledge, skills, and abilities necessary for success in future careers. The goal is to expose students to a wide array of career possibilities and to broaden their

future aspirations.

The CyberForce Competition is geared towards the successes and learning experiences of the competing collegiate student teams, but the event would not be possible without the extent of the wide array of volunteers involved. There are the technical roles of “hackers,” non-technical roles who assist with quality control and quality assurance, pre-competition documentation scoring, as well as logistical support. Volunteers come from all backgrounds and different professional avenues.

Structure

Throughout the event, there are various channels of interaction which facilitate the CyberForce Competition. It is also broken down into the various categorical scoring avenues listed below. Each year, a different energy-driven scenario is developed to reflect emerging trends and issues in both the energy sector and cybersecurity fields. Scenarios focus on various energy sector fields such as electricity, solar energy, wind energy, and hydropower, etc.

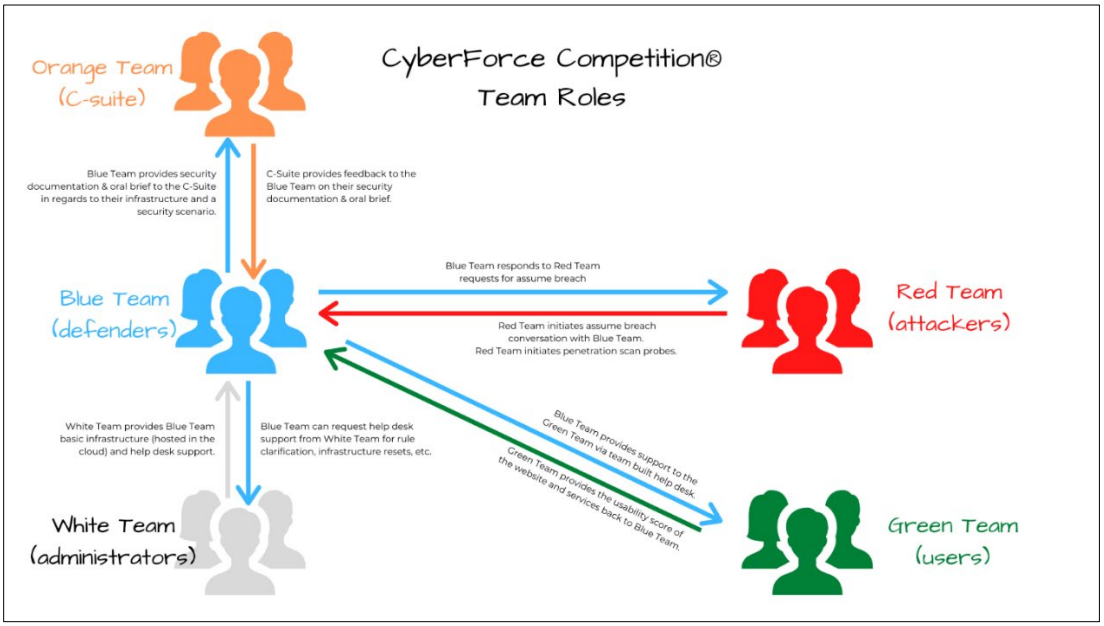
Communication Structure

In many competitions, there are color-coded teams. For the CyberForce Competition, we have the below outline of teams in which are regularly utilized in rules, talks, etc. when we speak about the benefits, communication, and overall play that occurs throughout the day.

Blue	A Blue team is composed of collegiate students who defend their network infrastructure from the Red team and maintain system usability for the Green team.
Red	The Red team includes industry security professionals that play the role of cyber attackers or “hackers,” attempting to breach the Blue network infrastructure and defenses of

	the Blue team participants.
Green	The Green team includes volunteers with a variety of skill sets, to emulate typical end users.
White	The White team includes national laboratory employees who support the participants in setting up their infrastructure and judge the competition.
Orange	The Orange team includes volunteers who play the role of a C-suite within a mock organization who review videos and security documentation.

Additionally, some teams interact more than others. The Blue team is the main team being that this competition is built for *them.* The below diagram highlights the general communication flow seen in the CyberForce Competition.



Teams

Blue Team

Blue team is comprised of collegiate students, ranging from Freshman to PhD, and are the defenders in the Competition setting. This is a team competition. Teams can range from 4-6 individuals, all

who must be currently attending the same institution at the time of the event. All educational backgrounds are encouraged to attend and are not limited to the degree students are seeking. Participants must be 18+ by the time of the event.

Red Team

Red team are volunteers from various industry, government, and national laboratories throughout the U.S. They are the “attackers” in the competition environment. They are responsible for the traditional attacks and the cyber infrastructure. There are also assumed breaches throughout the Competition. These simulate cyber incidents that have already occurred on systems, and are intended for teams to investigate, respond, and report the incidents.

Green Team

Green team represents the end user. The team consists of volunteers of various backgrounds, which may be technical or non-technical in abilities. Their main objective is to test the usability of Blue team websites and services throughout the duration of the competition. *For more detailed information, please see the Green 101 document.*

Orange Team

The Orange team emulates personnel who make up a company’s top management positions, referring to the C-Suite. Volunteers represent this mock organizational management, reviewing briefing videos and security documentation submitted by the Blue team. *For more detailed information, please see the Security Documentation and C-Suite Panel documents.*

Scoring

Blue Scoring

The Blue team scoring is based on the team's ability to keep services active and available ensuring, all while adhering to competition rules. Teams earn points for each availability scans that results in positive service uptime. They will also receive points for positive uptime of their industrial control system based on the scenario provided.

Red Scoring

This team's primary goal is to disrupt the confidentiality, integrity, and availability of the Blue Team networks. Additionally, they will be providing an educational platform for the students to learn about the attacks and threats found. Scoring is based upon assumed breach scenarios as well as penetration testing that occur throughout the entirety of the competition.

Orange Scoring

C-Suite Panel Brief and Security Documentation fall under the Orange scoring. These tasks are both completed pre-competition. Scoring is judged two weeks before the competition and reflects a team's ability to brief a mock leadership team based on a particular scenario. Documentation is also scored the week prior to the competition, and scores are given based on how well a Blue team can highlight their unique approaches to securing their infrastructure. Orange scores are a way to help competitors practice their soft skills of writing and public speaking.

Green Scoring

The Green team will review and complete surveys to evaluate each Blue team system's usability and user experience. This is the role of quality assurance and quality control.

Anomaly Scoring

Blue teams also can complete anomalies throughout the duration of the competition. Anomaly scoring is based on how many challenges teams answer correctly in the scoreboard. These range from time management challenges to energy and cybersecurity trivia, to cybersecurity skills. The challenges span from easy to hard, and points are awarded to correctly answered anomalies. Points are reflective of their challenge level. *For more detailed information, please see the Anomalies 101 document.*

References

- https://cyberforce.energy.gov/
- https://cyberforce.energy.gov/cyberforce-competition/
- https://cyberforce.energy.gov/cyberforce-competition/eligibility/
- https://cyberforce.energy.gov/cyberforce-competition/prior-competitions/